

Quantitative Skills

Question Bank

Questions and Answers

What is the Rational form of 99.172517251725....?

Solution:

Given that,

Number = 99.172517251725... (Here, the repeating numbers are 1725)

Let $x = 0.172517251725...$ (repeating 1725)

Multiply by 10,000 to shift one full repeating block:

$$10,000x = 1725.17251725...$$

Subtract the original x:

$$10,000x - x = 1725.1725... - 0.1725... = 1725$$

$$9,999x = 1725$$

Solve for x:

$$x = 1725 \div 9999$$

$$\text{Reduce by 3: } x = 575 \div 3333$$

Write the whole number:

$$99.1725... = 99 + x = 99 + 575/3333$$

Combine into a single fraction:

$$99 + 575/3333 = (99 \times 3333 + 575) \div 3333$$

$$= (329967 + 575) \div 3333$$

$$= 330542 \div 3333$$

Hence, the rational form = 330542/3333.

What is the Rational form of 88.142514251425....?

Solution:

Given that,

Number = 88.142514251425... (Here, the repeating numbers are 1425)

Let the repeating fractional part be

$$x = 0.142514251425\dots$$

Multiply by 10,000 to shift one full repeating block:

$$10,000x = 1425.14251425\dots$$

Subtract the original x:

$$10,000x - x = 1425.1425\dots - 0.1425\dots = 1425$$

$$9,999x = 1425$$

Solve for x:

$$x = 1425 \div 9999$$

$$\text{Reduce the fraction by 3} \rightarrow x = 475 \div 3333$$

Write the whole number with the fraction:

$$88.1425\dots = 88 + 475/3333$$

Combine into a single fraction:

$$(88 \times 3333 + 475) \div 3333 = (293304 + 475) \div 3333 = 293779 \div 3333$$

Hence, the rational form = $293779/3333$.

What is the Rational form of 22.172172172....?

Solution:

Given that,

Number = 22.172172172... (Here, the repeating numbers are 172)

Let the repeating fractional part be

$$x = 0.172172172\dots$$

Multiply by 1000 to shift one full repeating block:

$$1000x = 172.172172\dots$$

Subtract the original x:

$$1000x - x = 172.172\dots - 0.172\dots = 172$$

$$999x = 172$$

Solve for x:

$$x = 172 \div 999$$

Write the whole number with the fraction:

$$22.172172\dots = 22 + 172/999$$

Combine into a single fraction:

$$(22 \times 999 + 172) \div 999 = (21978 + 172) \div 999 = 22150 \div 999$$

Hence, the rational form = $22150/999$.

What is the remainder when 4^{60} is divided by 6?

Solution:

Given that,

We need the remainder when 4^{60} is divided by 6

Check the remainder cycle of 4 when divided by 6

$$4^1/6 = 4 \text{ remainder}$$

$$4^2/6 = 2 \text{ remainder}$$

$$4^3/6 = 4 \text{ remainder}$$

Here, the cycle starts repeating, so the cycle becomes {4,2}

Take the power value 60 and divide by 6, here, the remainder is 0

So, check for the remainder cycle. The 0 element is 4 here, so 4 is the remainder

What is the remainder when 7^{80} is divided by 10?

Solution:

Given that,

We need the remainder when 7^{80} is divided by 10

Check the remainder cycle of 7 when divided by 10

$$7^1/10 = 7 \text{ remainder}$$

$$7^2/10 = 49 \text{ remainder } 9$$

$$7^3/10 = 343 \text{ remainder } 3$$

$$7^4/10 = 2401 \text{ remainder } 1$$

Here, the cycle starts repeating, so the cycle becomes {7, 9, 3, 1}

Take the power value 80 and divide by 4 (cycle length), here the remainder is 0

So, check the remainder cycle. The 0 remainder corresponds to the 4th element of the cycle, which is 1, so 1 is the remainder

What is the remainder when 11^{100} is divided by 12?

Solution:

Given that,

We need the remainder when 11^{100} is divided by 12

Check the remainder cycle of 11 when divided by 12

$$11^1/12 = 11 \text{ remainder}$$

$$11^2/12 = 1 \text{ remainder}$$

$$11^3/12 = 11 \text{ remainder}$$

Here, the cycle starts repeating, so the cycle becomes {11, 1}

Take the power value 100 and divide by 2 (cycle length), here the remainder is 0

So, check the remainder cycle. The 0 remainder corresponds to the 2nd element of the cycle, which is 1, so 1 is the remainder

What is 20% of 25% of 300?

Solution:

Given that,

We need to find 20% of 25% of 300

As we know, a percentage can be applied step by step or multiplied directly:

$$20\% \text{ of } 25\% \text{ of } 300 = (20/100) \times (25/100) \times 300$$

Now, compute step by step,

$$25\% \text{ of } 300 = (25/100) \times 300 = 75$$

$$20\% \text{ of } 75 = (20/100) \times 75 = 15$$

Hence, the required value = 15.

What is 25% of 40% of 200?

Solution:

Given that,

We need to find 25% of 40% of 200

As we know, percentage can be applied step by step or multiplied directly:

$$25\% \text{ of } 40\% \text{ of } 200 = (25/100) \times (40/100) \times 200$$

Now, compute step by step,

$$40\% \text{ of } 200 = (40/100) \times 200 = 80$$

$$25\% \text{ of } 80 = (25/100) \times 80 = 20$$

Hence, the required value = 20.

What is the value of 140% of 56 + 56% of 140 ?

Solution:

Given that,

We need to evaluate: 140% of 56 + 56% of 140

As we know, $x\% \text{ of } y = (x/100) \times y$

Compute each part,

$$140\% \text{ of } 56 = (140/100) \times 56 = 1.4 \times 56 = 78.4$$

$$56\% \text{ of } 140 = (56/100) \times 140 = 0.56 \times 140 = 78.4$$

Now, add the results,

$$\text{Total} = 78.4 + 78.4 = 156.8$$

Hence, the required value = 156.8.

If the simple interest on Rs. 8700 for 2 years is Rs. 1740/-, what is the rate of interest percent per annum?

Solution:

Given that,

Principal amount = 8700

Time = 2 years

Simple interest = 1740

As we know, the formula to find simple interest is $SI = (P \times R \times T) / 100$

We need the rate of interest, so write the formula in terms of R,

$$R = (SI \times 100) / (P \times T)$$

Now, substitute the values in the above formula,

$$R = (1740 \times 100) / (8700 \times 2)$$

$$R = 174000 \div 17400$$

$$R = 10$$

Hence, the rate of interest per annum = 10%

If the simple interest on Rs. 4500 for 3 years is Rs.1350, what is the rate of interest per annum?

Solution:

Given that,

Principal amount = 4500

Time = 3 years

Simple interest = 1350

As we know, the formula to find simple interest is $SI = (P \times R \times T) / 100$

We need the rate of interest, so write the formula in terms of R,

$$R = (SI \times 100) / (P \times T)$$

Now, substitute the values in the above formula,

$$R = (1350 \times 100) / (4500 \times 3)$$

$$R = 135000 \div 13500$$

$$R = 10$$

Hence, the rate of interest per annum = 10%

If the simple interest on Rs. 4,000 for 3.5 years is Rs. 700, what is the rate per annum?

Solution:

Given that,

Principal amount = 4000

Time = 3.5 years

Simple interest = 700

As we know, the formula to find simple interest is $SI = (P \times R \times T) / 100$

We need the rate of interest, so write the formula in terms of R,

$$R = (SI \times 100) / (P \times T)$$

Now, substitute the values in the above formula,

$$R = (700 \times 100) / (4000 \times 3.5)$$

$$R = 70000 \div 14000$$

$$R = 5$$

Hence, the rate of interest per annum = 5%

Find the cost price of an object, when the selling price is Rs. 377 and the gain is 16%.

Solution:

Given that,

Selling price = 377

Gain = 16%

As we know, the relation between selling price and cost price is

$$SP = CP \times (100 + \text{Gain}\%) / 100$$

We need the cost price, so write the formula in terms of CP,

$$CP = SP \times 100 / (100 + \text{Gain}\%)$$

Now, substitute the values in the above formula,

$$CP = 377 \times 100 / 116$$

$$CP = 37700 \div 116$$

$$CP = 9425 \div 29$$

$$CP = 325$$

Hence, the cost price = Rs. 325.

If CP of 45 articles is equal to SP of 40 articles. Find the profit or loss percent.

Solution:

Given that,

Cost price of 45 articles = selling price of 40 articles

Let CP per article = C and SP per article = S

From the given relation: $45C = 40S$

Express S in terms of C: $S = (45/40) \times C = (9/8) \times C$

We know that, Profit percent = $[(S - C) / C] \times 100$

Substitute $S = (9/8)C$ in above formula,

$$\text{Profit percent} = [(9/8)C - C] / C \times 100 = (1/8) \times 100$$

$$\text{Profit percent} = 12.5\%$$

Hence, there is a profit of 12.5%.

The cost price of 48 muffins equals the selling price of 60 muffins. Find the profit or loss percent.

Solution:

Given that,

Cost price of 48 muffins = selling price of 60 muffins

Let CP per muffin = C and SP per muffin = S

From the given relation: $48C = 60S$

Express S in terms of C:

$$S = (48/60) \times C = (4/5)C$$

We know that, Profit or loss percent = $[(S - C) / C] \times 100$

Substitute $S = (4/5)C$ in above formula,

$$[(4/5)C - C] / C \times 100 = [-1/5] \times 100 = -20$$

Hence, there is a loss of 20%.

The ratio of two numbers is 2:3. If the HCF of those two numbers is 9, find the LCM of those two numbers.

Solution:

Given that,

Ratio of the two numbers = 2 : 3

HCF of the two numbers = 9

Let the numbers be $2k$ and $3k$

Since $\text{HCF}(2k, 3k) = k$ and HCF is 9, we get $k = 9$

Numbers = $2 \times 9 = 18$ and $3 \times 9 = 27$

Use the relation: $a \times b = \text{HCF} \times \text{LCM}$

So, $18 \times 27 = 9 \times \text{LCM}$

Compute the product: $18 \times 27 = 486$

Now, $\text{LCM} = 486 \div 9 = 54$

Hence, the LCM of the two numbers = 54.

Find the LCM and HCF of 12, 20 & 45?

Solution:

Given that,

Numbers = 12, 20, 45

As we know, prime factorization helps to find HCF and LCM

$$12 = 2 \times 2 \times 3 = 2^2 \times 3$$

$$20 = 2 \times 2 \times 5 = 2^2 \times 5$$

$$45 = 3 \times 3 \times 5 = 3^2 \times 5$$

HCF takes the common prime factors with the smallest powers across all three

Common primes across all three numbers = none
So, HCF = 1

LCM takes all prime factors with the highest powers present

Pick 2^2 (from 12 or 20), 3^2 (from 45), and 5 (from 20 or 45)
 $LCM = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$

Hence, HCF = 1 and LCM = 180.

Find the sum of the factors of 36.

Solution:

Given that,

Number = 36

Prime factorization of $36 = 2^2 \times 3^2$

As we know, the sum of all factors for $n = a^p \times b^q$ is
 $S = ((a^{p+1} - 1) / (a - 1)) \times ((b^{q+1} - 1) / (b - 1))$

Here $a = 2$, $p = 2$ and $b = 3$, $q = 2$

Compute each term,

For 2^2 :

$$(2^{(2+1)} - 1) / (2 - 1) = (2^3 - 1) / 1 = (8 - 1) / 1 = 7$$

For 3^2 :

$$(3^{(2+1)} - 1) / (3 - 1) = (3^3 - 1) / 2 = (27 - 1) / 2 = 26 / 2 = 13$$

Now, multiply the two terms,

$$\text{Total sum of factors} = 7 \times 13 = 91$$

Hence, the sum of all factors of $36 = 91$.

A painter is hired to paint a community hall. The estimated requirement of paint is 20 kg. Accounting for 10% extra for wastage, and knowing that the paint is sold in 2 kg cans at Rs.16 per can, what will be the total cost of paint purchased?

Solution:

Given that,

Estimated paint required = 20 kg

Extra for wastage = 10% of 20 kg = $0.10 \times 20 = 2$ kg

Total paint needed = $20 + 2 = 22$ kg

Paint is sold in 2 kg cans

Number of cans required = $22 \div 2 = 11$ cans

Cost per can = Rs. 16

Total cost = $11 \times 16 = \text{Rs. } 176$

Hence, the total cost of paint purchased = Rs. 176.

At Sunrise High School, there are 480 students enrolled this year. The principal shared that 60% of the students are boys, while the rest are girls. Based on this information, find the number of boys and girls studying in the school.

Solution:

Given that,

Total students = 480

Percentage of boys = 60%

Percentage of girls = $100\% - 60\% = 40\%$

Number of boys = 60% of 480 = $(60/100) \times 480 = 0.6 \times 480 = 288$

Number of girls = 40% of 480 = $(40/100) \times 480 = 0.4 \times 480 = 192$

Hence, boys = 288 and girls = 192.

Three traffic lights at a busy intersection change simultaneously. The first light changes every 5 seconds, the second every 8 seconds, and the third every 10 seconds. If they all start together, how many times will they change at the same time in 20 minutes?

Solution:

Given that,

First light changes every 5 seconds

Second light changes every 8 seconds

Third light changes every 10 seconds

Total time = 20 minutes = $20 \times 60 = 1200$ seconds

As we know, they change together every LCM of (5, 8, 10) seconds

Prime factors: $5 = 5$, $8 = 2^3$, $10 = 2 \times 5$

$\text{LCM} = 2^3 \times 5 = 8 \times 5 = 40$ seconds

Number of simultaneous changes in 1200 seconds (including the start)

$$= (1200 \div 40) + 1$$

$$= 30 + 1$$

$$= 31$$

Hence, they will change together 31 times in 20 minutes.

In a park, two fountains spray water at regular intervals. The first fountain sprays every 12 minutes, and the second fountain sprays every 18 minutes. If both fountains spray water at the same time at 9:00 AM, after how many minutes will they spray together again?

Solution:

Given that,

First fountain interval = 12 minutes

Second fountain interval = 18 minutes

As we know, they will spray together every LCM of the two intervals

Prime factors: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$

$\text{LCM} = 2^2 \times 3^2 = 4 \times 9 = 36$ minutes

Hence, they will spray together again after 36 minutes.

A bakery earned a total profit of Rs. 2700 in a week. The owner decides to share the profit among three staff members in the ratio of 2 : 3 : 4 according to their contribution. How much will the first staff member receive?

Solution:

Given that,

Total profit = 2700

Sharing ratio = 2 : 3 : 4

Total parts = 2 + 3 + 4 = 9

Amount per part = $2700 \div 9 = 300$

First staff member's share = 2 parts = $2 \times 300 = 600$

Hence, the first staff member will receive Rs. 600.

There are three equal glasses filled with milk and water. Each glass contains the following proportion of milk to water, first glass 1:2, in the second as 3 : 4 and in the third 5 :2, The contents of the three glasses are emptied into a single vessel. What is the proportion of milk and water in the new vessel?

Solution:

Given that,

Three equal glasses are mixed together, with milk : water ratios:

First glass = 1 : 2

Second glass = 3 : 4

Third glass = 5 : 2

Let each glass contain 1 unit of mixture (equal volumes assumed)

Milk and water fractions in each glass:

First glass $\rightarrow$ milk = $1/(1+2) = 1/3$, water = $2/3$

Second glass $\rightarrow$ milk = $3/(3+4) = 3/7$, water = $4/7$

Third glass $\rightarrow$ milk = $5/(5+2) = 5/7$, water = $2/7$

Total milk in vessel = $1/3 + 3/7 + 5/7 = 1/3 + 8/7$

Convert to common denominator 21 $\rightarrow 1/3 = 7/21, 8/7 = 24/21$

Total milk = $7/21 + 24/21 = 31/21$

Total water in vessel = $2/3 + 4/7 + 2/7 = 2/3 + 6/7$

Convert to common denominator 21 $\rightarrow 2/3 = 14/21, 6/7 = 18/21$

Total water = $14/21 + 18/21 = 32/21$

Proportion of milk : water = $(31/21) : (32/21) = 31 : 32$

Hence, the proportion of milk and water in the new vessel = 31 : 32.

A shopkeeper wants to divide his total profit of Rs. 2700 among his three employees in the ratio 2 : 3 : 4 based on their performance. What will be the share of the first employee?

Solution:

Given that,

Total profit = 2700

Sharing ratio = 2 : 3 : 4

Total parts = $2 + 3 + 4 = 9$

Amount per part = $2700 \div 9 = 300$

First employee's share = 2 parts = $2 \times 300 = 600$

Hence, the first employee will receive Rs. 600.

PART B

Section A - Multiple Choice Questions (MCQs)

1. Which of the following is a prime number?

- A) 21
- B) 17
- C) 25
- D) 27

Answer: B) 17

Solution: A Prime number is a number greater than 1 that is divisible only by 1 and itself.

- 21 is divisible by 3 and 7, which is not prime
- 17 is divisible only by 1 and 17 which is prime
- 25 is divisible by 5 which is not prime
- 27 is divisible by 3 which is not prime

2. The HCF of 16 and 24 is:

- A) 6
- B) 8
- C) 12
- D) 16

Answer: B) 8

Solution: HCF is the greatest number that divides both numbers.

- Factors of 16: 1, 2, 4, 8, 16
- Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
- Common factors: 1, 2, 4, 8

So, $\text{HCF} = 8$

3. The LCM of 5 and 7 is:

- A) 35
- B) 12
- C) 25
- D) 42

Answer: A) 35

Solution:

LCM is the smallest number divisible by both numbers.

$$5 \times 7 = 35$$

So, $\text{LCM} = 35$

4. Which of the following is a factor of 30?

- A) 7
- B) 5
- C) 8
- D) 11

Answer: B) 5

Solution: $30/5 = 6$

So, 5 is a factor of 30

5. 15% of 200 is:

- A) 20
- B) 25
- C) 30
- D) 35

Answer: C) 30

Solution:

$$15\% \text{ of } 200 = (15 \times 200)/100 = 3000/100 = 30$$

6. If the cost price of an item is Rs. 150 and selling price is Rs. 180, what is the profit?

- A) Rs. 20
- B) Rs. 25
- C) Rs. 30
- D) Rs. 35

Answer: C) Rs. 30

Solution: Profit = SP – CP = 180 – 150 = 30

7. Loss occurs when:

- A) SP > CP
- B) SP < CP
- C) SP = CP
- D) None of these

Answer: B) SP < CP

Solution: Loss occurs if selling price is less than cost price.

8. A product costs Rs. 600. If there is a 10% discount, what is the selling price?

- A) Rs. 540
- B) Rs. 560
- C) Rs. 580
- D) Rs. 600

Answer: A) Rs. 540

Solution: Discount = 10% of 600 = 60

$$SP = 600 - 60 = 540$$

9. The unit digit of 43×27 is:

- A) 1
- B) 9
- C) 3
- D) 7

Answer: A) 1

Solution:

As we need to find Unit digit only ,

$$3 \times 7 = 21$$

So, unit digit = 1

10. Which of the following is a multiple of 9?

- A) 27
- B) 28
- C) 31
- D) 35

Answer: A) 27

Solution: $27 / 9 = 3$

3 is divisible by 9

11. Which of the following numbers is an even number?

- A) 21
- B) 34
- C) 47
- D) 53

Answer: B) 34

Solution: Even numbers divisible by 2

Here, $34 / 2 = 17$

12. If a number increases from 50 to 55, what is the percentage increase?

- A) 5%
- B) 10%
- C) 15%
- D) 20%

Answer: B) 10%

Solution: Increase = $55 - 50 = 5$

% Increase = $(5/50) \times 100 = 10\%$

13. What is the smallest number divisible by both 4 and 10?

- A) 20
- B) 30
- C) 40
- D) 50

Answer: A) 20

Solution: LCM of 4 and 10 = 20

So, 20 is the smallest number divisible by both

14. Which number is a factor of 60?

- A) 7
- B) 10
- C) 13
- D) 11

Answer: B) 10

Solution: $60/10 = 6$

So, 10 is a factor of 60

15. Calculate 10% of 250.

- A) 20
- B) 25
- C) 30
- D) 35

Answer: B) 25

Solution: $10\% \text{ of } 250 = (10 \times 250)/100 = 25$

16. If the cost price of an item is Rs. 120 and selling price is Rs. 150, what is the profit?

- A) Rs. 20
- B) Rs. 25
- C) Rs. 30
- D) Rs. 35

Answer: C) Rs. 30

Solution: Profit = SP – CP = 150 – 120 = 30

17. When the selling price equals the cost price, the seller:

- A) Makes a profit
- B) Makes a loss
- C) Neither makes profit nor loss
- D) Cannot be determined

Answer: C) Neither makes profit nor loss

Solution: SP = CP → no profit, no loss

18. An item priced at Rs. 900 is sold with a 20% discount. What is the selling price?

- A) Rs. 720
- B) Rs. 750
- C) Rs. 780
- D) Rs. 800

Answer: A) Rs. 720

Solution: Discount = 20% of 900 = 180

$$SP = 900 - 180 = 720$$

19. What is the unit digit of the product 23×17 ?

- A) 1
- B) 3
- C) 7
- D) 9

Answer: A) 1

Solution: Unit digit only

$$\text{So, } 3 \times 7 = 21$$

So, unit digit = 1

20. Which of the following numbers is divisible by 4?

- A) 14
- B) 18
- C) 20
- D) 22

Answer: C) 20

Solution: $20/4 = 5$

20 is divisible by 4

Section B – Fill in the Blanks

1. The sum of all prime numbers between 2 and 10 is ____

Answer: 17

Solution: Primes between 2 and 10 $\rightarrow 2, 3, 5, 7 \rightarrow \text{sum} = 2 + 3 + 5 + 7 = 17$

2. The number of zeros at the end of the product $75 \times 95 \times 77$ is ____

Answer: 0

Solution: Trailing zeros come from factors of 10 $\rightarrow$ factor of 2×5 .

- $75 = 5 \times 5 \times 3 \rightarrow \text{no } 2$

- $95 = 5 \times 19 \rightarrow \text{no } 2$

- $77 = 7 \times 11 \rightarrow \text{no } 2$

As we have no factor of 2, So there will be no trailing zeros

3. The LCM of 26 & 52 is ____

Answer: 52

Solution: 52 is a multiple of 26

LCM = 52

4. 2% of 2 = ____

Answer: 0.04

Solution: $2 \times 2/100 = 0.04$

5. The formula for Profit Percentage is ____

Answer: $(\text{Profit} / \text{Cost Price}) \times 100$

6. The formula for Loss Percentage is ____

Answer: $(\text{Loss} / \text{Cost Price}) \times 100$

7. A farmer has 12 apples and wants to divide them between his two children in the ratio 1:2. The first child will get ____ apples

Answer: 4

Solution: Total parts = $1 + 2 = 3$

first child = $12 \times \frac{1}{3} = 4$ apples

8. A baker has 18 cupcakes and wants to divide them between two friends in the ratio 1:2. The first friend will get ____ cupcakes

Answer: 6

Solution: Total parts = 3

first friend = $18 \times \frac{1}{3} = 6$ Cup cakes

9. A shirt costs Rs. 500. If there is a 10% discount, the price of the shirt after the discount is ____

Answer: 450

Solution: Discount = 10% of 500 = 50

SP = $500 - 50 = 450$

10. A textbook is listed at ₹600. If there is a 10% discount, the price of the textbook after the discount is ____

Answer: 540

Solution: Discount = 10% of 600 = 60

SP = $600 - 60 = 540$

11. HCF stands for ____ Common Factor

Answer: Highest

12. In the formula $SI = (P \times T \times R) / 100$, P stands for ____

Answer: Principal

13. The value of the expression $(8! \times 7!) / (6! \times 5!)$ is ____

Answer: 2352

Solution:

- $8! \times 7! / (6! \times 5!) = (8 \times 7 \times 6!) \times (7 \times 6 \times 5!) / (6! \times 5!)$
- Cancel 6! and 5! $\rightarrow (8 \times 7) \times (7 \times 6) = 56 \times 42 = 2352$

14. LCM stands for ____ Common Factor

Answer: Least

15. In the formula $SI = (P \times T \times R) / 100$, T stands for ____

Answer: Time